Learning is acquiring new knowledge, behaviours, skills, values, preferences or understanding, and may involve synthesising and processing different types of information.

Cognitive psychology is the study of how people perceive, learn, remember, and think about information.

Cognitive – to recall, calculate, discuss, analyse, problem solve, etc.;

Memory is the process by which we encode, store, and retrieve information."

Sensory memory is the brief storage of information in its original sensory form, lasting only a fraction of a second.

Short-term memory is a limited-capacity store that can hold information for a few seconds to half a minute.

Long-term memory is the relatively permanent and unlimited storehouse of the memory system.

MEMORY PROCESSES:

Memory processes can be understood in three stages. They can be easily memorized as 3R's of memory (register, retain, retrieve):

Encoding (or registration)

Encoding is the process of converting the sensory information coming from the environment that can be registered in the memory system.

Storage (retention)

Storage is the process of converting the incoming information into permanent records so that it can be accessed later when needed.

Retrieval (recall, or recognition)

Retrieval is the process of locating and leveling the stored information from the storehouse to the consciousness at the time of need.

MEMORY MODELS

Multi-Store Model (Atkinson & Shiffrin, 1968): Memory consists of three sequential stores—sensory memory, short-term memory (STM), and long-term memory (LTM). Information passes from sensory input to STM through attention, and can be transferred to LTM through rehearsal.

Working Memory Model (Baddeley & Hitch, 1974): STM is an active system for holding and manipulating information, consisting of a central executive, phonological loop, visuospatial sketchpad, and episodic buffer for integrating information.

Levels of Processing Model (Craik & Lockhart, 1972): Memory retention depends on the **depth of processing**; deeper, meaningful processing and elaborative rehearsal result in stronger, long-term memory compared to shallow processing.

Parallel Distributed Processing / Connectionist Model: Memory is stored as patterns of activation across interconnected nodes, with parallel processing and distributed representation, and learning occurs by adjusting connection strengths.

Dual-Store Models of LTM: Long-term memory is divided into **explicit (declarative)** memory for conscious recall and **implicit (non-declarative)** memory for unconscious skills and habits.

Personality is the dynamic organization within the individual of those psychophysical systems that determine his characteristic behavior and thought.

PERSONALITY THEORIES

Psychoanalytic Theory of Personality (Sigmund Freud)

Personality develops through unconscious sexual and aggressive drives in conflict with societal rules, shaped largely in early childhood, and grows as impulses are channeled into socially acceptable behaviors.

Allport's Trait Theory

Personality consists of consistent traits, with common traits shared by society and personal traits unique to the individual, shaped mainly through socialization and remaining stable across situations.

Cattell's Trait Theory

Personality is structured by surface traits (observable) and source traits (underlying), influenced by genetics and environment, and includes dynamic, ability, and temperamental traits that motivate and shape behavior.

Carl Rogers: Theory of Personality

Personality develops within an individual's perceived reality, guided by an innate actualizing tendency, where congruence between experiences and self promotes personal growth and adaptive behavior.

EYSENCK'S PERSONALITY THEORY: THE THREE-FACTOR MODEL

Eysenck's three-factor (PEN) model views personality as biologically based, consisting of **Extraversion, Neuroticism, and Psychoticism**, organized hierarchically from broad traits to specific behaviors.

Extraversion: Extraversion reflects sociability and energy; extraverts seek stimulation, while introverts prefer calm. It is linked to cortical arousal, with introverts naturally more aroused than extraverts.

Neuroticism: Neuroticism indicates emotional instability; highly neurotic individuals are anxious and reactive, while stable individuals handle stress calmly, influenced by autonomic nervous system responsiveness.

Psychoticism: Psychoticism reflects traits such as aggressiveness, impulsivity, and tough-mindedness, with higher levels linked to creativity or nonconformity but also potential antisocial tendencies.

MEASUREMENT OF PERSONALITY

Subjective Methods: Subjective methods of measuring personality rely on personal judgment, introspection, and opinions rather than standardized scoring.

Objective Methods: Objective methods use standardized tools and structured approaches to assess personality, providing measurable and replicable results.

Projective Methods: Projective methods assess personality indirectly by presenting ambiguous stimuli, based on the idea that individuals project their inner thoughts, feelings, and conflicts onto the task.

Intelligence: According to American Psychology Association (APA, 1995), Individuals differ from one another in their ability to understand complex ideas, to adapt effectively to the environment, to learn from experience, to engage in various forms of reasoning, to overcome obstacles by taking thought.

Theories

Spearman's Theory of Intelligence: Charles Spearman's two-factor theory posits that intelligence is comprised of a general factor (g-factor), which influences performance across various cognitive abilities, and specific factors (s-factors), which are unique to particular mental activities. The g-factor is responsible for acquiring knowledge, abstract reasoning, and adapting to new situations.

Thurstone's Theory of Intelligence: L.L. Thurstone proposed that intelligence consists of seven Primary Mental Abilities (PMA), rather than a single general factor. These abilities include word fluency, verbal comprehension, spatial visualization, perceptual speed, numerical facility, reasoning, and associative memory. While initially thought to be independent, these abilities were later found to be intercorrelated, supporting Spearman's idea of a 'g' factor.

Sternberg's Theory of Intelligence: Robert Sternberg's Triarchic theory of intelligence describes three distinct types of intelligence: contextual (practical) intelligence, which is the ability to adapt to environmental demands; creative (experiential) intelligence, which involves developing new ideas and solutions; and analytical (componential) intelligence, which is measured by traditional IQ tests and involves abstract thinking and appraisal of situations.

Cattell's Theory of Intelligence

Raymond Cattell (1963) proposed the notion of fluid and crystallized intelligence.

Fluid intelligence is the capacity to reason and solve new problems. There is no influence from any knowledge from past experiences, rather the person innovates new logical methods to resolve the problem.

Crystallized intelligence is the ability to use skills, knowledge and past experiences.

PASS Theory: The Planning, Attention, Simultaneous, and Successive (PASS) Theory, developed by Das, Naglieri, and Kirby, outlines four key cognitive competencies. These include planning (making decisions and monitoring feedback), attention (selecting stimuli and ignoring others), simultaneous processing (integrating stimuli into a whole), and successive processing (integrating stimuli in a specific serial order). This model helps understand various cognitive processes such as reasoning, imagery, language, and memory

Gardner's Theory of Intelligence

Howard Gardener (1993b, 1999a) refuted the classic view of intelligence as a capacity for logical reasoning. He proposed that there is no one form of intelligence but a number of intelligence work together.

According to him, **intelligence** is the "ability to solve problems or fashion products that are of consequence in a particular cultural setting or community

Emotional Intelligence

The term "emotional intelligence" was coined by researchers Peter Salovey and John D. Mayer in 1990 in their article "Emotional Intelligence".
Peter Salovey and John Mayer, defined it as The ability to monitor one's own and other people's emotions, to discriminate between different emotions and label them appropriately, and to use this information to guide one's thinking and behavior

The intelligence quotient

The abbreviation "IQ" was coined by the psychologist William Stern for the German term Intelligenzquotient, his term for a scoring method for intelligence tests at University of Breslau he advocated in a 1912 book. Intelligence quotient (IQ) refers to mental age (MA) expressed as a ratio of chronological age (CA) multiplied by 100. For IQ to remain stable, MA must increase with CA over time. This is true until around 18 years, when intellectual abilities are usually fully developed.

MEASUREMENT OF INTELLIGENCE

Verbal assessment: Verbal intelligence assessments evaluate a person's language skills through tasks like vocabulary, reading comprehension, verbal reasoning, and general knowledge.

Nonverbal assessment: Nonverbal assessments evaluate problem-solving, logical reasoning, and spatial abilities using visual patterns, diagrams, and symbols instead of language.

Performance assessment: A performance assessment evaluates an individual's knowledge and skills through real-world tasks or product creation, emphasizing practical application over recall.

The **Culture Fair Intelligence Test (CFIT)** is a nonverbal test developed by Cattell to measure fluid intelligence, focusing on abstract and analytical reasoning with minimal cultural bias.

Culture-free test a test designed to eliminate cultural bias completely by constructing questions that contain either no environmental influences or no environmental influences that reflect any specific culture.

Hypothesis: A hypothesis is a tentative proposition suggested as a solution to a problem or as an explanation of some phenomenon.

Null Hypothesis (H_0) : The hypothesis that there is no relationship or no difference.

Alternative Hypothesis (H_1): The hypothesis that states the expected relationship or difference.

VARIABLE: A variable is a property that takes on different values.

Independent Variable (IV): "The variable manipulated or selected by the researcher to determine its effect on another variable."

Dependent Variable (DV): "The variable on which the effect of the independent variable is observed."

Extraneous Variable: "Variables that are not directly related to the purpose of the study but may affect the dependent variable."

Reliability: "Reliability refers to the consistency or stability of a measurement instrument."

Test–Retest Reliability: "Test–retest reliability is the degree to which scores are consistent over time."

Inter-Rater Reliability: "Inter-rater reliability is the extent to which different raters agree in their assessments."

Parallel-Forms Reliability: "Parallel-forms reliability is the consistency of two equivalent forms of a test."

Internal Consistency Reliability: "Internal consistency is the degree to which items within a test measure the same construct."

Validity: “Validity is the degree to which a test measures what it claims to measure.”

Content Validity: “Content validity is the extent to which a test represents the content domain it intends to cover.”

Criterion-Related Validity: “Criterion-related validity is the degree to which test scores relate to an external criterion.”

Construct Validity: “Construct validity is the extent to which a test measures the theoretical construct it is intended to measure.”